INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
TUTORIAL 6

1. A particle moves along a straight line under the influence of a deceleration of $\frac{3}{2}v^{2/3}$ where v is the velocity at time t from its origin A. If the particle moves with velocity u at A, show that it will stop at a distance of $\frac{1}{2}\ u^{4/3}$ in time $2u^{1/3}$.

2. An object of mass $m\ kg$ is given an upward velocity of $16\ ms^{-1}$ from rest, $16\ m$ above the ground. Assume that gravity is the only force acting on it, when will the object hit the ground and at what velocity will it hit the ground? (Let gravity, $g\ = 10\ ms^{-2}$).

3. An object of mass $m\ kg$ is given an upward velocity of $\frac{ng}{k}\ ms^{-1}$ and then allowed to fall under the influence of gravity. Assume the force in Newton (N) due to the air resistance is mkv, where v is the velocity of the object in ms^{-1}, and n and k are constants. Show that the object reaches its maximum height after $\frac{1}{k}\ln(n+1)$ and its maximum height is $\frac{g}{k^2}(\mathrm{n} - \ln(n+1))$.

4. A parachutist whose weight is $75\ kg$ drops from a helicopter hovering $2000\ m$ above the ground and falls towards the ground under the influence of gravity. Assume that the force due to the air resistance is proportional to the velocity of the parachutist, with the proportionality constant $k_1 = 30\ kg\ s^{-1}$ when the chute is close and $k_2 = 90\ kg\ s^{-1}$ when the chute is open. If the chute does not open until the velocity of the parachutist reaches $20\ ms^{-1}$, after how many seconds will it be before the parachutist reach the ground? (Let $g = 9.81\ ms^{-2}$.)

5. A particle of mass m which moves along a horizontal straight line with a velocity of v encounters a resistance of $av + bv^3$, where a and b are constants. If there is no other force beside the resistance acting on the particle, and its initial velocity is U, show that the velocity is $\frac{1}{2}U$ after time $\frac{m}{2a}ln\left[\frac{4a+bU^2}{a+bU^2}\right]$.